

DATABASE SPECIFICATIONS

*Myki Card Transportation System Database
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Revision Sheet

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Rev. 0	26/03	Complete Chapter 1 – General Information
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1.0 GENERAL INFORMATION

GENERAL INFORMATION

1.1 Purpose

This document defines the logical and physical design specifications for the **Myki Card Transportation System Database**. The primary purpose of this specification is to replicate the core functionality of the existing Myki system while focusing on optimising its backend database performance, including:

- **Touch-on / touch-off trip recording**
- Fare calculation and capping rules
- Transaction tracking (fare deductions and top-ups)
- Vehicle and station tracking for determining travel locations
- System extensibility for various transport modes (tram, train, bus)

This project aims to address the key database performance bottlenecks identified in the current system by applying advanced database design principles, such as index tuning, partitioning strategies, and effective storage management. Additionally, this specification estimates long-term storage requirements for the next 10 years based on projected usage growth and data accumulation patterns.

The proposed database model is designed for implementation in a Microsoft SQL Server environment and serves as a reference for database administrators, system developers, and other stakeholders involved in its development and maintenance.

1.2 Scope

This database specification defines the design and structure of the backend relational data storage for the Myki Card System, with a primary focus on supporting public transport operations within the metropolitan area for full fare passengers.

In particular, this specification covers:

- Modelling of Myki cards and customer account information
- Capturing touch-on and touch-off events through scanning devices
- Recording trip histories along with relevant travel metadata
- Storing transaction records for fare deductions, top-ups, and Myki Pass purchases
- Integrating vehicles, routes, and stop stations to determine precise device location data

- Deriving boarding and alighting stations based on vehicle movement logs and GPS tracking data

The scope of this specification explicitly excludes regional transport logic, special fare concession handling (e.g., child, concession, senior fares), and any integration with external fare enforcement or financial settlement systems. These components may be considered for future system enhancements.

This defined scope provides a robust foundation for capturing and processing core Myki usage scenarios while ensuring the database design is extensible for future operational requirements.

1.3 System Overview

This system represents a simplified smartcard fare collection platform similar to the Myki card system used in Victoria's metropolitan public transport network. The primary objective of this system is to serve as a proof-of-concept for implementing advanced relational database design and optimisation techniques in a public transport setting.

The database design emphasises key operations like touch-on/touch-off event processing, trip logging, fare calculation, transaction recording, and vehicle location data integration. This system creates a structured environment for showcasing database modelling principles, performance tuning methods, and storage optimisation techniques.

- Responsible organisation:

Not Applicable—This system was developed as part of an academic coursework project within RMIT University for educational purposes. No formal organisation is responsible for production operations.

- System name or title:

Myki Card Transportation System Database

- System code:

Not Applicable

- System category:

Major application — The system performs defined functions (touch-on, trip tracking, fare deduction) with identifiable logical components and business rules. This system is not a general-purpose support system.

- Operational status:

Under Development — This system is in the design and prototyping phase for academic demonstration and is not intended for production deployment.

- System environment and special conditions:

This database system is developed within a controlled educational environment. Key assumptions and design conditions include:

- Target Relational Database Management System (RDBMS): Microsoft SQL Server
- Assumes integration with simulated GPS and log-based inputs for vehicle movement tracking
- Focus is limited to backend database design; no front-end interface, API integration, or external payment processing components are within the scope of this system.

1.4 Project References

The following external resources were used to inform and support the design of this database specification:

1. **Public Transport Victoria (PTV) GTFS Feed – Vehicle Types**
URL: <https://data.ptv.vic.gov.au/downloads/GTFSReleaseNotes.pdf>
2. **Train Service Passenger Counts – Data Structure Reference**
URL: <https://discover.data.vic.gov.au/dataset/train-service-passenger-counts/resource/162887ef-1dba-4d9b-83bd-baee229229c6>
3. **TramTracks Real-Time Tram Location Feed**
URL: <https://discover.data.vic.gov.au/dataset/tram-tracks/resource/60dc379e-1ac9-4e3c-9dca-c428f5c014f0>
4. **Myki Fare Structure – Metropolitan Zones and Default Fares**
URL: <https://www.ptv.vic.gov.au/tickets/fares/metropolitan-fares/#defaultfares>
5. **Victoria Population projection:**
URL: https://www.planning.vic.gov.au/_data/assets/pdf_file/0022/703453/DTP0552-Victori-in-Future-2023-report.PDF
6. **Method of travel to work:**
<https://www.abs.gov.au/census/guide-census-data/census-dictionary/2021/variables-topic/transport/method-travel-work-mtwp>
7. **PTV Annual report:**
<https://www.vic.gov.au/dtp-annual-report>

1.5 Acronyms and Abbreviations

Acronyms or Abbreviations	Description
DB	Database
DBMS	Database Management System
RDBMS	Relational Database Management System
PTV	Public Transport Victoria
GPS	Global Positioning System
SQL	Structured Query Language
T-SQL	Transact-SQL (Microsoft SQL Server extension to SQL)
ERD	Entity-Relationship Diagram
RMIT	Royal Melbourne Institute of Technology

1.6 Points of Contact

1.6.1 Information

Name: Jordan Chiou

Role in this project: Database Designer

Contact: jungdechiou@gmail.com / s4068959@student.rmit.edu.au

1.6.2 Coordination

Not Applicable — No additional coordination roles are defined for this project.

1.6.3 Additional Points of Contact

Not Applicable — No other contributors or stakeholders are involved in this project.

1.6.4 Data Owners

Data Owner	Responsibility
Jordan Chiou	Responsible for database design, data modelling, and overall data ownership for the project.

2.0 DATABASE IDENTIFICATION AND DESCRIPTION

DATABASE IDENTIFICATION AND DESCRIPTION

2.1 Naming Conventions

1. Logical Naming Conventions:

Logical names are used in entity-relationship (ER) diagrams and conceptual discussions.

- **Entities** (tables) use **PascalCase**, singular form.
 - Example: MykiCard, Trip, VehicleRun, StopStation
- **Attributes** (columns) use **snake_case**, descriptive and self-explanatory.
 - Example: card_id, touch_on_time, vehicle_type
- Primary keys are named as entity_name_id or abbreviated as xxx_id followed with {PK} symbol.
 - Example: trip_id {PK}, route_id{PK}
- Foreign keys are not explicitly marked in this ER diagram but are named identically to the referenced primary key.
 - Example: card_id in Trip references MykiCard.card_id
- Enumerated values (e.g. CardType, VehicleType) are shown as fixed values or look-up tables.

2. Physical Naming Conventions (SQL Server)

These conventions apply to the actual implementation of the schema in SQL Server.

- **Table names** follow the same PascalCase convention, plural form is **not** used.
 - Example: VehicleRun, CustomerAccount
- **Column names** use lowercase with underscores (snake_case), consistent with the logical model.
- **Constraints and indexes** use descriptive names with standard prefixes:
- **Data types** are chosen based on usage:

2.2 Database Identification

The database defined in this specification is referred to as:

| MykiTransportDB

This name uniquely identifies the logical schema of the database used to simulate the operation of a Mykicard-based fare collection system for a metropolitan public transport network.

All data entities (tables) within this database follow a clear and consistent naming convention as defined in Section 2.1. Each table name serves as its unique identifier, without the use of prefixes or abbreviations, to promote readability and traceability.

Database Code Name

- **Logical Database Name:** MykiTransportDB
- **Schema (if applicable):** myki

Table Identifier:

Table Name	Description
CustomerAccount	Stores user information for registered customers
MykiCard	Represents each issued card, balance, and type
Trip	Captures travel journeys initiated by card taps
Transaction	Logs monetary operations such as fare deductions or top-ups
Scanner	Represents touch-on/off scanning devices
DeviceLocation	Abstract link between scanner and physical location
StopStations	Fixed locations where passengers board or alight
Route	Defines public transport routes
RouteStop	Mapping of ordered stop points for each route
Vehicle	Transport units (tram, bus, train)
VehicleType	Lookup for types of vehicles
VehicleRun	A scheduled or real-time instance of a vehicle operating a route
VehicleStopLog	System-generated record of vehicle arrival/departure at stations
VehicleRealTimeLog	Raw GPS location log for tracking vehicle movement
CardType	Enumerated card fare types (full fare, child, etc.)

All table names are unique and self-descriptive within the scope of the MykiTransportDB schema.

2.3 Systems Using the Database

Not applicable

2.4 Relationship to Other Databases

Not applicable

2.5 Schema Information

2.5.1 Description

The database is implemented as a single schema named **MykiTransportDB**, representing a modular and normalised relational structure. This schema encompasses all entity, reference, and operational tables required to simulate a Mykicard-based fare management system.

Schema Name

- Logical schema name: **MykiTransportDB**
- Target RDBMS: **Microsoft SQL Server**

Sub-schemas

- Not Applicable — the system is implemented as a single schema without physical sub-schemas.

File Type / Data Model

- Relational model
- All tables are defined using **T-SQL (Transact-SQL)**
- Primary and foreign key constraints are used to enforce referential integrity.

Partitioning Strategy

To support scalability and query performance for high-volume operational data, horizontal partitioning is proposed for the following tables. The partition key is based on datetime fields, enabling range-based partitioning by month.

Table	Partition Key	Partition Strategy
Trip	touch_on_time	Monthly range partition
Transaction	timestamp	Monthly range partition
VehicleStopLog	actual_arrival_time	Monthly range partition
VehicleRealTimeLog	timestamp	Monthly range partition

Indexing Strategy

Indexing is applied to improve lookup performance and enforce foreign key constraints. Below are the proposed indexes:

Table	Indexed Columns	Purpose
Trip	card_id, touch_on_time	Fast card-based and time-based queries
Transaction	card_id, trip_id, timestamp	Historical filtering and joins
VehicleStopLog	run_id, stop_station_id	Match vehicle logs
VehicleRealTimeLog	vehicle_id, timestamp	Real-time location lookup
Scanner	device_location	Resolve scanner placement quickly
MykiCard	customer_id	Filtering cardholders

Access Control Keys

- Not Applicable — no user-level security is implemented in this version.

Concurrency and Locking

- Default SQL Server concurrency mechanisms apply.
- No custom concurrency control or isolation level overrides are implemented.

Data Name Mapping

- All table and column names follow clear and consistent naming conventions as described in Section 2.1.
- No aliases, synonyms, or abstraction layers are used.

Partition and File Limitations

- No table or index is horizontally sharded across multiple filegroups.
- Entire schema is designed to operate within a single logical database instance.
- Partitioned tables are logically segmented for performance, but physical filegroup optimization is out of scope.

Access Path Restrictions

- No materialized views, full-text indexes, or computed columns are defined.
- All data access relies on defined primary keys and basic indexing.

2.5.2 Physical Design

Graphically depict the physical design of the database.

2.5.3 Physical Structure

Describe and depict in a graphic representation the physical structure (partitions, files, indexes, pointers) and the logical components of the database. Identify the criteria required to achieve operating efficiency.

2.6 Data Dictionary

Reference the data dictionary and attach it as an appendix to this document.

2.7 Special Instructions

Identify instructions to be followed by personnel who will contribute to the generation of the database and who will use it for testing and operational purposes. Such instructions may include:

- Identify any specialized criteria for entering data into the database.
- Identify source documents for the rules and procedures to be followed when submitting data for entry into the database.
- Identify source documents for the machine run instructions for generating, modifying, updating, or otherwise using the database. In very large systems, in which the details of such instructions are extensive, reference sections of other documents in which this specific information may be found.

3.0 DATABASE ADMINISTRATIVE INFORMATION

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3.1 Responsibility

Identify the organizations and personnel responsible for the following database administrative functions: database administrator, system administrator, and security administrator. Describe specific administration skill requirements.

3.2 System Information

Document the Database Management System configuration, hardware configuration, database software utilities, and any support software used.

3.2.1 Database Management System (DBMS) Configuration

Identify the vendor, version or release date and targeted hardware for the DBMS. Describe any restrictions on the initialization and use of the DBMS to support any intended distributed processing.

3.2.2 Hardware Configuration

Identify the hardware configurations on which the database will reside.

3.2.3 Database Software Utilities

List and reference the documentation of any DBMS utility software available to support the use or maintenance of the database.

3.2.4 Support Software Available for Maintaining Database

Describe all support software, including the operating system, directly related to the database, including name, version, function, and major operating characteristics. Cite documentation by title, number, and appropriate sections. Examples of such software include database management systems, query language, report writers, storage allocation software, database loading software programs, and file processing programs, and data cleaning software.

3.2.5 Security

Describe the use and management of integrity and access controls that apply to all database components such as schema, sub-schema, partitions or physical files, records or tables, sets or relations, and data elements.

3.3 Storage Requirements

Describe the storage device. Provide sizing formulas for determining the storage required to support the database content and associated software. Estimate the internal and peripheral storage requirements. Identify multiple storage requirements for distributed processing.

3.4 Recovery

Describe the methodology for reestablishment or recreation of the necessary data schema and system support files.

3.5 Partition/File Information

3.5.1 Content

Describe the content of each partition/file, listing the records it contains and explaining the purpose.

3.5.2 Description

Describe the design and format of each partition/file, including name, type, code, mapping, limitations and controls, access procedures, and mechanisms.

3.5.3 Partition/File Interdependencies

Identify the interdependencies of each partition/file in the database.

3.6 Database Interfaces

Provide a description of the interfaces with other application software including these of other operational capabilities and from other organizations. For each interface, specify the following information:

3.6.1 Description of Operational Implications

Describe operational implications of data transfer, including security considerations.

3.6.2 Description of Data Transfer Requirements

Describe data transfer requirements to and from the software, including data content, format, sequence, and any conversion issues.

3.6.3 Description of Formats of Data

Describe formats of data for both the sending and receiving systems, including the data item names, codes, or abbreviations that are to be interchanged, as well as any units of measure/conversion issues.

3.7 Error Handling

Describe those system error handling routines and procedures that are available during execution of database software.